

Section: **3.6 – BOLTED JOINTS IN COMPOSITES**
Chapter: **3610**
Title: **RIVETS LOADS**
Revision: **1.0**
Date: **29/04/2020**

1 INTRODUCTION

This chapter extends the applicability of chapters §3210 to §3215 about the methodologies applicable to calculate fasteners loads for the different types of mechanically fastened joints between structural components, when at least one of them is made of fibrous composite materials.

The following considerations shall have to be taken into account in such case.

2 RIVETS LOADS IN ATTACHMENTS TO COMPOSITE PARTS

2.1 STIFFNESS OF A FASTENER

Stiffness of a Fastener in Shear Load

The Huth's formula, as proposed in chapter §3210 "TYPES OF RIVETED JOINTS", sub-section 2.1, shall also be used for bolted joints of plates made of composite materials. Also for hybrid joints, with a plate produced from a metallic alloy and another from a composite laminate.

$$C = \left(\frac{t_1 + t_2}{2d} \right)^a \frac{b}{n} \left(\frac{1}{t_1 E_1} + \frac{1}{nt_2 E_2} + \frac{1}{2t_1 E_3} + \frac{1}{2nt_2 E_3} \right)$$

Group I—bolted metallic joints	$a = 2/3; b = 3.0$
Group II—riveted metallic joints	$a = 2/5; b = 2.2$
Group III—bolted graphite/epoxy joints	$a = 2/3; b = 4.2$
(*) Hybrid metallic/CFRP bolted joints	$a = 2/3; b = 3.6$

Figure 1. Huth's formula for fastener flexibility, plus the case with hybrid joints (*)

Remarks:

- ✓ For hybrid joints in which one plate is made of a metallic alloy and another in a composite CFRP material, the simple average of the parameters proposed by Huth for metallic and CFRP joints is suggested here (*)
- ✓ In case the direction of the rivet load is known a-priori, the apparent elastic modulus of the laminate in that direction shall be used in the formula (see chapter §3401 "STRESS AND STRAIN IN COMPOSITE LAMINATES" for further support)
- ✓ In case the direction of the rivet load is NOT known a-priori, the elastic modulus to use in the formula shall be that of the quasi-isotropic laminate made with the same material, noted below as E_{QI} . That value can be calculated from the laminate engineering constants E_x , E_y , G_{xy} , ν_{xy} as follows...

Non zero terms of the reduced in-plane stiffness matrix of the laminate, assuming laminate follows standard design rules, and no shear-extension coupling occurs, are:

$$Q_{11} = E_x / (1 - \nu_{xy} \cdot \nu_{yx}) \quad \dots \quad \nu_{yx} = \nu_{xy} \cdot E_y / E_x$$

$$Q_{22} = E_y / (1 - \nu_{xy} \cdot \nu_{yx})$$

$$Q_{12} = \nu_{xy} \cdot E_y / (1 - \nu_{xy} \cdot \nu_{yx})$$

$$Q_{21} = Q_{12}$$

$$Q_{66} = G_{xy}$$

A quasi-isotropic laminate can be built with a stacking sequence including a 25% of the plies aligned with 0° material direction, 50% of the plies at ±45° directions and the remaining 25% at 90° direction. Taking advantage of formulation presented in Jones (Ref. 1), pages 72, 77, ...

$$Q_{11}^{QI} = 0.25 \cdot Q_{11} + 0.25 \cdot Q_{22} + 0.5 \cdot 0.25 \cdot [Q_{11} + 2 \cdot (Q_{12} + 2 \cdot Q_{66}) + Q_{22}]$$

$$Q_{12}^{QI} = 0.5 \cdot Q_{12} + 0.5 \cdot [0.25 \cdot (Q_{11} + Q_{22} - 4 \cdot Q_{66}) + 0.5 \cdot Q_{12}]$$

$$\nu_{QI} = Q_{12}^{QI} / Q_{11}^{QI}$$

$$E_{QI} = Q_{11}^{QI} \cdot (1 - \nu_{QI}^2)$$

Axial Stiffness of a Fastener in Tension Load

Methodology proposed in chapter §3210 "TYPES OF RIVETED JOINTS", sub-section 2.2 shall be used. For composite plates with bolts in tension, the elastic modulus of the base laminate in the stiffness formula shall be replaced by the applicable flexural elastic modulus of the laminate, which can be obtained from the inverted bending stiffness matrix of the laminate as explained in chapter §3401 "STRESS AND STRAIN IN COMPOSITE LAMINATES":

$$E_f = 12 / (t^3 \cdot [D^{-1}]_{ii})$$

The sub-index ii denotes the term of the inverted matrix consistent with the direction facing the wall/leg closest to the tensioned bolt in the base plate. Eg. if the wall/leg closest to the bolt is parallel to the Y-axis of the base laminate, the ii index shall be 11.

2.2 TYPES OF RIVETED JOINTS

The description of the types of fastened joints, as well as the recommended methodologies for calculating their loads distributions, as included in chapters §3210 to §3215, are still applicable if any one of the members attached is made of composite materials.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
CFRP	-	Carbon Fibre Reinforced Polymer
E_f	MPa	Flexural Young's modulus of the laminate
E_x	MPa	Apparent elastic modulus of the laminate in X direction
E_y	MPa	Apparent elastic modulus of the laminate in Y direction
$[D]$	Nmm	Bending stiffness matrix of the laminate
G_{xy}	MPa	Apparent shear elastic modulus of the laminate in its plane
ν_{xy}	-	Poisson's ratio of the laminate
QI	-	Quasi-isotropic laminate
Q_{ij}	MPa	Terms of the reduced in-plane stiffness matrix of the laminate ($[Q] = [A] / t$)
t	mm	Thickness of laminate
TBC	-	To Be Confirmed

REFERENCES

<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1 [Robert M. Jones]	2 nd Ed.	MECHANICS OF COMPOSITE MATERIALS

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures (module JOINTSTIFF)

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub- chapters affected
1.0 29/04/2020	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Methods (TEASP-TL3)	G. Baños 19/03/2020	F. Glock 29/04/2020

Table 3. *Approval signatures table for last revision of this chapter*